Research Statement – Sean Patrick MacBride

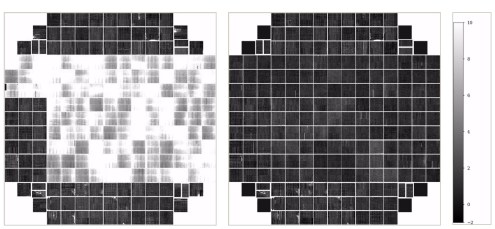
My research focuses on exploring the cosmic frontier using astronomical instrumentation, multi-messenger astrophysics, and standard siren cosmology. In this proposal, I will describe how my past experience and leadership across multiple collaborations has prepared me to continue exploring the frontier as a KICP Fellow at the University of Chicago.

Early Career Contributions: My previous research has focused on instrumentation development for ground based optical telescopes, target-of-opportunity searches for electromagnetic counterparts of gravitational wave events, and constraining dark energy using electromagnetic and gravitational wave data.

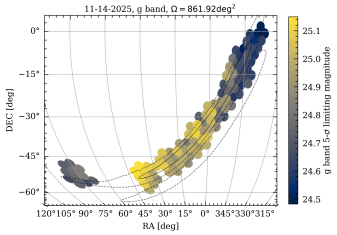
Astronomical Instrumentation Development: My recent work has spanned both photometric and spectroscopic instruments. This began in 2020, with the **Large Lenslet Array Magellan Spectrograph** (LLAMAS), where I validated the optical performance of the diffraction gratings and collimators for the LLAMAS spectrographs, assembled and verified optical tolerance for the 24 spectrograph cameras, and integrated all 2,400 AR-coated fibers into the focal plane with zero fiber loss, before it was commissioned on the Magellan Baade telescope in 2024 (1; 2). After integrating and testing LLAMAS, I shifted my focus from integral-field spectrographs to multi-object spectrographs. Starting in 2023, as a PhD student at the University of Michigan-Ann Arbor, I tested a common failure mode on **Dark Energy Spectroscopic Instrument** (DESI) robotic positioners, where I found that reorienting the robotic positioner to an inverted configuration could restore nominal behavior of the robotic positioners. This indicated that gravity-sag has a significant influence on the presence and recovery of these failures (3).

This expertise with spectroscopic instruments translated effectively to the transcendent photometric imager of the next decade, the **LSST Camera** (LSSTCam). Shortly after beginning my PhD at the University of Zürich, I moved to Chile in 2024 to serve as an LSSTCam Summit Support Scientist, where **I led LSSTCam focal-plane commissioning** on Cerro Pachón as a principal component of the in-kind contribution from Switzerland to the Rubin Project. In this role, I tested the CCDs in both the summit clean-room and on the Simonyi Survey Telescope, and optimized the sensor performance. This characterization and testing played a leading role in achieving LSSTCam first photon in April 2025 (see Figure 1a), culminating in an invited proceeding to the SPIE Astronomical Telescopes + Instrumentation conference in July 2026, an impactful astronomical instrumentation conference held biennially (4).

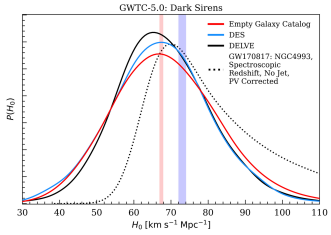
Target-of-Opportunity Searches: Since 2023, I have led twenty observing campaigns using different telescopes and facilities, with the goal of detecting electromagnetic (EM) counterparts to gravitational wave (GW) merger events. A variety of multi-messenger analyses are enabled by the detection of an EM counterpart to a GW merger event, and my particular focus is to combine the redshift information from the host galaxy of an EM counterpart with the luminosity distance inferred from the GW source. With this combination, it is possible to constrain the local rate of accelerated expansion of the universe, known as the Hubble Constant (H_0) in the standard model of cosmology, ΛCDM. This technique is known as the *bright siren* method (7).



(a) Removal of the persistence effect in LSSTCam CCDs, a dominant, time-dependent, residual charge effect (4).



(b) The deepest wide area ToO search to date, for GW candidate S251112cm, using Vera C. Rubin Observatory (5).



(c) Constraints on H_0 using the fifth Gravitational Wave Transient Catalogue and the DELVE DR3 galaxy catalog (6).

Figure 1: A summary of early career contributions, spanning instrumentation, multi-messenger astrophysics, and standard siren cosmology.

Since 2023, I have have been co-leading the Dark Energy Survey Gravitational Wave follow-up program (DESGW) program, conducting **Dark Energy Camera** (DECam) observations, processing raw image data from sixteen Target-of-Opportunity (ToO) campaigns to calibrated products, and analyzing processed data in response to LIGO-Virgo-Kagra (LVK) Observing Run 4 (O4) triggers. During a recent observing campaign in 2025, I led a collaboration between DESGW and the Japanese collaboration for gravitational-wave electromagnetic follow-up (J-GEM) when a well-localized gravitational wave merger candidate was accessible to both Northern and Southern hemisphere instruments. In this collaboration, I obtained multi-epoch photometry using DECam, processed the obtained data, and cross-matched this data with spectroscopic observations using **Prime Focus Spectrograph** (PFS), a recently commissioned multi-object spectrograph at the Subaru Telescope. This analysis enabled the first scientific result using PFS (8).

Since 2025, I have been leading the Vera C. Rubin Observatory (Rubin) ToO observing program as the Rubin ToO observer. In this role, I have led observing responses to one interstellar object (9), one high-energy neutrino event (later reclassified as terrestrial), and two gravitational-wave merger events, including the deepest, wide-area GW ToO search to date ($m_{g,5\sigma} \sim 24.8$ across $\sim 850\,\mathrm{deg}^2$, see Figure 1b) (5; 10). The process to build, test, and deploy the automated system for Rubin is reflected in the recent proceedings of SPIE Astronomical Telescopes + Instrumentation (5), closely following the community-defined science case for the program (10).

Standard Siren Constraints on the Hubble Constant: While recent observing campaigns have not yielded an EM counterpart to a GW merger, I have retained a focus on constraining cosmology using EM and GW data, employing the *dark siren* method to measure the local expansion rate of the universe, H_0. The dark sirens method infers redshift information of GW events using a catalog of potential host galaxies, which is lower precision than the bright siren method, but can be applied to all GW events (11). This began by supporting the preparation of Dark Energy Survey (DES) data, resulting in the first dark siren constraint using a deep photometric catalog to jointly infer GW population parameters and cosmological parameters (12). DES was accepted as the fiducial result for the most recent LVK analysis, and demonstrated a 22% improvement in constraining H_0 compared to previous analyses (13). I have since expanded on the DES preparation, leading a dark-siren analysis using DECam Local Volume Exploration (DELVE), where we show that when photometric redshift biases are properly treated, **DELVE outperforms DES using dark sirens** (see Figure 1c), constraining H_0 to 11% when combined with the bright siren GW170817 (6).

Future Research Activities: My proposed research builds on this foundation through complementary directions. First, I will characterize image sensors for future cosmic surveys, while also deploying monitoring tools for the LSST Camera to improve LSST data quality. Second, I will lead Target-of-Opportunity searches to detect the next bright siren and other exotic astrophysical events. Third, I will usher standard siren cosmology into a precision era, leveraging LSST data alongside specialized observational programs for bright and dark siren cosmology.

Instrumentation Development for Future Cosmic Surveys: Building on my experience commissioning and testing CCDs for the LSST Camera focal plane, I propose to characterize the next generation of astronomical sensors, focusing on the multi-amplifier sensing (MAS) CCDs slated for **DESI Run 2** and the **Stage-5 Spectroscopic Experiment** (Spec-S5). This work will apply my expertise LSSTCam characterization to test, qualify, and optimize MAS CCDs for forthcoming spectroscopic surveys. In doing so, I will ensure that the detector performance meets the underlying requirements of DESI Run 2 and Spec-S5, maintaining the precision cosmology goals from the project's inception. This testing campaign will have impacts in the near-term, as DESI Run 2 is expected to start in 2029, and will pathfind for a longer time-horizon experiment, Spec-S5.

In addition to testing image sensors for future astronomical instruments, I will maintain a support-scientist role for LSSTCam as the Legacy Survey of Space and Time (LSST) transitions into survey operations, maintaining the operational knowledge needed to respond quickly to sensor anomalies and focal-plane issues as they arise. Expanding on my commissioning-era characterization of LSSTCam defects, I will also deploy a defect-monitoring tool, currently in-development, to `cp_verify`, the calibration-product verification framework used by the LSST Science Pipelines (14). This tool will elevate defect monitoring to an automated process, improving LSST data quality.

Bright Siren Searches Enabled by Rubin Observatory: I will lead the observing planning, execution, and analysis of the next bright standard siren accessible to Rubin Observatory, continuing my role as the Rubin ToO observer during the transition to operations. While it is difficult to predict the timing of the next bright siren, I will continue to deploy improvements to the ToO system to improve response latency and observing strategy, maximizing the next opportunity to detect and characterize these optical counterparts and obtain a well-constrained measure of H_0. In parallel to operating the Rubin ToO system, I will analyze DESGW candidates identified during future LVK observing runs. This positions me to sustain and support Rubin's ToO program with additional imaging in other filters and extend the temporal-sampling using a similarly capable photometric imager. This complimentary support will play an important role, especially as other priorities of the LSST are balanced against the ToO program. As the LSST ToO program matures, I will work to enhance transient characterization by securing spectroscopic follow-up time on other instruments, moving beyond photometric classification to confirm spectral features that can inform the scope of Rubin's ToOs, and constrain the physical properties of LSST transients.

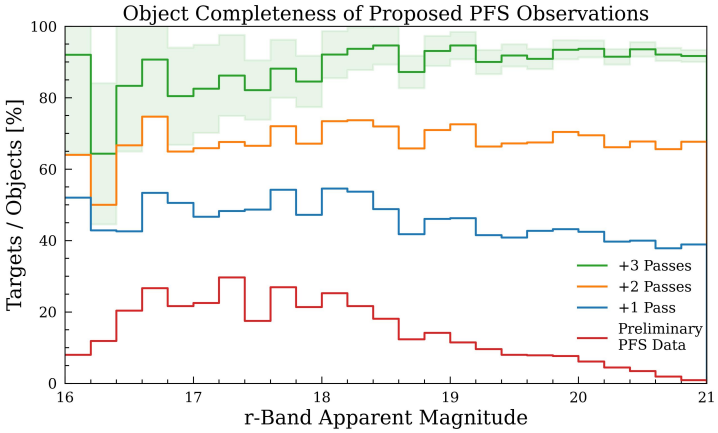


Figure 2: In red, the completeness of PFS objects acquired as part of our ToO campaign (8), with other colors depicting the forecasted completeness of proposed observations. The high completeness obtained over the S250328ae localization region in 3 passes (one night) demonstrates the utility of multi-object spectrographs as characterization machines for standard siren cosmology and transient astronomy.

Dark Siren Measurements for Precision Cosmology: I will pursue a cutting-edge program that creates a gold-standard photometric galaxy catalog for dark siren cosmology using LSST data. This program is structured in four stages, each building towards a powerful, world leading measurement. It begins with an ongoing project, focused on characterizing the LSST Data Preview 2 (DP2) galaxy catalog, aiming to define robust and scalable object classification for future LSST data releases. Following characterization of DP2, I will finish a forecasting study that quantifies the H_0 precision achievable by combining simulated first-year LSST data with an O4-like gravitational-wave catalog, extending the dark-siren methodology from my recent DELVE measurement to project the constraining power in the LSST era. Superceding DP2 preparation, I will characterize the first LSST Data Release (DR1) galaxy catalog and produce a dark siren measurement of H_0. This result will extend the methodology I developed with DELVE to a dataset with substantially greater depth, area, and photometric uniformity, all features that will make it the standard galaxy catalog for dark siren analyses. Building on the DR1 measurement, I aim to combine LSST large-scale structure measurements in a joint cosmological analysis, using cosmic shear and standard sirens to obtain a tighter constraint of H_0, and extend the constraints to Ω_m.

To assess the frontier of the dark siren method using targeted observations, I will lead a dedicated spectroscopic dark-siren pilot survey, obtaining targeted redshifts within GW localization volumes to assess the limits and enhancements to H_0 constraints using well-localized dark sirens. This survey is led through a continuing collaboration with J-GEM on the PFS instrument, obtaining additional observations for the GW merger pursued in our collaborative DESGW ToO with an expected observation date starting in April 2027 (see Figure 2).

As a KICP Fellow, I will achieve these research goals by collaborating with fellow detector specialists, including Alex Drlica-Wagner, to characterize instrumentation for future spectroscopic instruments. In addition to my instrumentation focus, I will collaborate with Daniel Holz and other extragalactic transient experts to search and characterize the next electromagnetic counterpart of a gravitational wave event using Rubin Observatory. Through this concerted effort, I will propel advancements in astronomical instrumentation, push the boundaries of multi-messenger astrophysics, and mature standard sirens, both bright and dark, into a precision cosmological probe.

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